DroneHacks 2026 — Navigation Approaches

Team

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1 Overview

Each approach gets its own file in `approaches/`: the idea, where it lives in `refnav/`, its assumptions, and what testing in the sim showed. Add a row to the table when an approach is added or tested.

Approach	Code	Status
Open-space explore	`main`	Implemented, not flown
Hardcoded route	`hardcoded-route-draft`	Draft, unverified
Wall vault	`hardcoded-route-draft`	Draft, untested

2 Open-space explore

Idea. No map. After climbing, repeatedly steer toward the most open direction in the depth image.

How. `Mission._explore` in `refnav/core.py`, enabled with `"explore": true` in `route.json`. Each depth frame scores `explore_bearings` headings across $\pm$`explore_half_fov_deg`:

$$\text{score}(b) = \min(\text{clearance}(b),\ \text{explore_range_m}) - \text{explore_straight_bias} \cdot |b|$$

and turns toward the best one. Lands after `explore_s`.

Assumptions / risks. Only sees inside the camera's field of view, so it can loop or get stuck in dead ends. Has no notion of a goal.

Sim results. None yet.

3 Hardcoded route

Idea. Plan offline from the course CAD and fly fixed waypoints.

How. On branch `hardcoded-route-draft` (Aryan). Waypoints were found with A* over an occupancy grid built from `dronehacks-course-v3.obj`, walls inflated by 30 cm. 20 waypoints from the launch mark to the landing pad through Gate B; the window bonus is not used. Plotted in `diagrams/route-plan.html`.

Assumptions / risks.

- The launch heading faces $+Y$ in the CAD frame. Unverified; if wrong, every waypoint is rotated.

- The sim maze matches the CAD. Unverified.
- 30 cm margin vs. the real airframe span. Unverified.
- Open loop: any VIO drift accumulates over the whole route.

Sim results. None yet.

4 Wall vault

Idea. When blocked, climb over the obstacle instead of landing.

How. On branch `hardcoded-route-draft` (Rohan), `Mission._vault`. Phases: approach to `vault_standoff_m`, climb until depth no longer sees the wall (capped at `vault_max_alt_m`), fly `vault_overstep_m` forward, then land. Triggered by `vault_on_obstacle` or `vault_only`.

Assumptions / risks.

- Must be allowed by the course rules.
- With `vault_on_obstacle`, any obstacle on the route triggers it, not only walls meant to be crossed.
- It ends the mission (lands) rather than rejoining the route.

Sim results. None yet.